

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

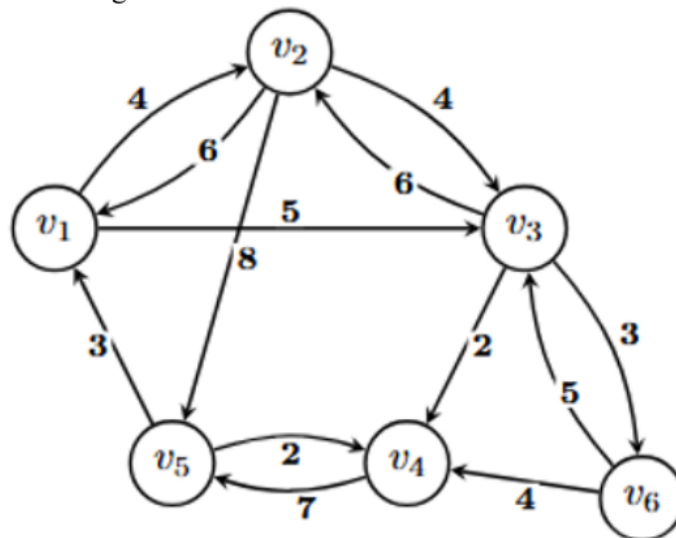
Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which pair of conditions defines a complement b of a in a Boolean algebra? (1)
(A) $a \vee b = 0, a \wedge b = 1$
(B) $a \vee b = 1, a \wedge b = 0$
(C) $a \vee b = a, a \wedge b = b$
(D) $a = b$
- 2) How many complements does each element of a Boolean algebra have? (1)
(A) None
(B) At least two
(C) Infinitely many
(D) Exactly one
- 3) Which expression equals $a \vee (\neg a \wedge b)$? (1)
(A) $a \vee b$
(B) $a \wedge b$
(C) $\neg a \vee b$
(D) a
- 4) Which expression is in disjunctive normal form (a disjunction of conjunctions of literals)? (1)
(A) $(x \vee y) \wedge z$
(B) $\neg(x \vee y)$
(C) $(x \wedge y) \vee (\neg x \wedge z)$
(D) $(x \vee y) \wedge (\neg x \vee z)$
- 5) Which expression is in conjunctive normal form (a conjunction of disjunctions of literals)? (1)
(A) $(x \wedge y) \vee z$
(B) $(x \vee y) \wedge (\neg x \vee z)$
(C) $\neg(x \wedge y)$
(D) $(x \wedge y) \vee (\neg x \wedge z)$

Section B: Theory questions (25 marks)

- 6) Draw the Hasse diagram for $A = \{1, 2, 3, 12, 18, 36\}$, ordered by divisibility: a is related to b iff a divides b . Is it a lattice? Justify. List all antichains of length more than one. (5)
- 7) How many ways are there to select a total of 15 items from three types, subject to these limits? (4)
Type A: 1 to 4 items; Type B: 3 to 7 items; Type C: 5 to 10 items.
- 8) Show that the number of partitions of n with no part smaller than 3 equals the number of partitions of $n+3$ whose three largest parts are consecutive integers. (3)
- 9) Find the 75th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order. (3)
- 10) Show that if G is a regular self-complementary graph of order n , then $n = 4t + 1$ for some integer t . (3)
- 11) If G has p vertices and minimum degree $\delta(G) \geq (p - 1)/2$, prove that G is connected. (3)
- 12) Use Dijkstra's algorithm to find the shortest paths from v_1 to every other vertex in the directed network below. Show the steps. (4)



End of question paper